

# Design And Simulation Of a Wearable Body Area Antenna Cloud-Based Data Storage

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## *Abstract*

- In this project we aim to design a wearable antenna for telemedicine applications with antenna frequency between 4.17 – 4.25 GHz, covering the wide frequency range between 1.80 GHz to 4.87 GHz at this range of frequencies that are used for wireless telemedicine, M-health, E-health applications.
- Body-centric radio-frequency identification (RFID) technology provides a wireless and maintenance free communication link between the human body and the surroundings through wearable and implanted antennas.
- This enables real-time monitoring of human vital signs anywhere as with we record and store this acquired data from the designed antenna to cloud server.
- This enables remote access of the data along with providing security, automatic backup of data and collaboration with multi sources.

# Problem Statement

- Existing monitoring systems are **bulky, hospital-dependent, and unsuitable** for long-term real-time use.
- Current wearable antennas face challenges:
  - **High SAR** → unsafe for prolonged usage.
  - **Low efficiency** and **narrow bandwidth** → poor performance.
  - **Degradation near human tissue** due to lossy environment.
- Lack of **cloud integration** restricts remote access and live monitoring.
- Need for a **low-SAR, high-performance wearable antenna** that enables **safe operation, real-time data acquisition, cloud storage, and remote health monitoring**.
- The current system does not offer data security and wireless data access.
- Design and simulate a wearable antenna optimized for biomedical applications.

# *Objectives of the Proposal*

- Ensure:
  - Low SAR ( $< 1.6 \text{ W/kg}$ )
  - High gain & efficiency
  - Flexible and comfortable integration into clothing
- Enable data collection of vitals: heart rate, respiration, blood pressure, clot detection and also storing of important medical data and history of the patient .
- Store data in a secure cloud server for real-time remote monitoring and enabling data acquiring from any location and time with help of unique identification number .
- Wireless Body Area Networks(WBANs) enable communication between wearable biomedical devices. Wearable antennas are the vital components for Body Centric Communication (BCC)

# Introduction



- They are fundamental in the Wireless Body Area Networks (WBANs) for health care, military, sports, and identification purposes
- Their performance in terms of return loss, gain, directivity, bandwidth, radiation pattern, efficiency, and Specific Absorption Rate (SAR) is influenced by the coupling and absorption of the human body tissues .
- In most cases, these antennas are integrated into the clothes, or in some cases, they may be fixed over the skin of the users.



*Fields in which Wireless Body Area Networks is applicable*

# Literature Survey

## ❖ Design, Analysis and Applications of Wearable Antennas,IEEE(2023):

- ✓ The paper discusses the brilliance of wearable antennas integrated with metamaterial structures composed of wearable/flexible substrate materials, with a focus on single and dual band antenna designs.
- ✓ The paper also reviews the critical design issues, various fabrication techniques, and other factors that need to be considered in the design of wearable/textile/flexible antennas.
- ✓ It was studied how the antenna and the human body interact, the voxel and equivalent tissue models for the on-body analysis were presented, and the antenna performance on various human body parts was examined.
- ✓ It was found that the complex permittivity and conductivity of the lossy human body has an impact on the performance of the antenna, such as radiation pattern, resonant frequency, bandwidth, and efficiency in particular.
- ✓ In view of the mentioned facts, it was concluded that metamaterial-based antennas are good candidates for wearable applications.

❖ **Design And Analysis of Werable Antenna For Wireless Body Area Network,Conference on Frontiers in Materials and Smart System Technologies(2022):**

- ✓ This paper is a study of wearable antenna design for wireless body area network applications.
- ✓ The antenna is designed using ANSYS High Frequency Structural Simulator (HFSS) software.
- ✓ The designed antenna works at operating frequency of 2.45 GHz.
- ✓ A rectangular patch made up of copper is used as radiating patch and the ground is also made with copper. The substrate material used is bed sheet cotton with a relative permittivity of 3.27 and the loss tangent is 0.00786. The size of the antenna is 40mm\*34mm\*1.26mm. The properties such as gain directivity and VSWR are studied for the designed antenna. The properties are studied using the ANSYS HFSS software.
- ✓ This antenna provides a wider bandwidth with in the frequency range of 2.1GHz to 2.7GHz and 3.6 GHz to 4.3GHz.The antenna has an efficiency of 94%.
- ✓ The SAR value of the designed antenna is 1.25 which is less than the SAR limitation so this antenna will not cause radiation problem when placed on a human body.

❖ **On-Bod Miniature Implantable and Wearable y Antennas: Towards the New Era of Wireless Body-Centric Systems, IEEE Antennas and Propagation Magazine, Vol. 56, No. 1, February 2014:**

- ✓ This paper studied the design and characterization of antennas for body-centric sensing systems, with an emphasis on in-body RFID and on-body UHF RFID applications.
- ✓ It discussed the design of antennas for a wireless brain-machine interface system, considering the system's requirements and limitations.
- ✓ It provide a direct wireless communication path between the brain and an external device, wireless brain machine interface systems which will allow patients suffering from spinal-cord injury and other severely disabling neurological conditions to communicate and live everyday lives more independently.
- ✓ They have designed planar cm-sized loop antenna configurations as external antennas, and mm-sized three-dimensional loop antennas as the neural-sensor antennas.
- ✓ In this paper, a detailed review of the wireless powering of mm-size neural sensors through an inductive link is presented.

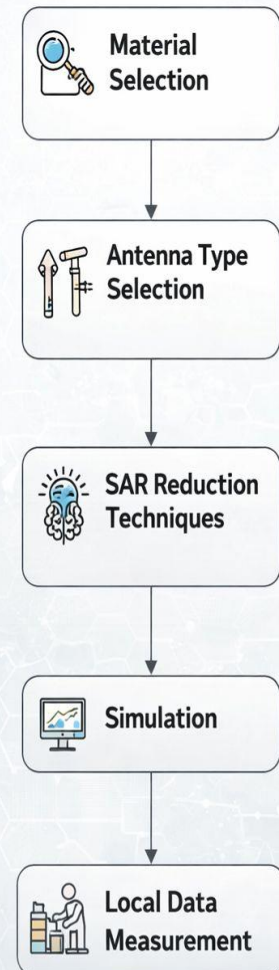
### ❖ *Flexible and Wearable Antenna for Biomedical Application: Progress and Opportunity,IEEE(2023):*

- ✓ While designing flexible antenna for wearable and biomedical applications, the conformal nature, low loss, and better flexibility plays a vital role to obtain high gain and low SAR value.
- ✓ A detailed study of several materials such as denim, cotton jeans, felt and polyester with low tangent loss and better flexibility were suggested in this study. This material also offered wide bandwidth, high gain, and better SAR reduction. SAR reduction (>90%) and high gain (~7.5 dB) were obtained using flexible material such as leather, denim, and cotton jean. Different non-rigid conductive materials were studied where copper adhesive, zelt, shield .It and conductive ink were suggested due to their effective flexibility.
- ✓ Here, coplanar waveguide feed is suggested with AMC ground to improve the gain and SAR reduction. Different metamaterial structures such as AMC, EBG and DGS were studied, and comparison was presented based on detailed study of gain and bandwidth improvement techniques.

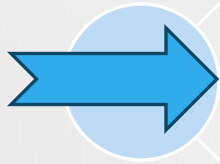
# Existing Methodologies

- ✓ Focus mainly on **antenna design only**.
- ✓ Use of **textile and flexible substrates** with conductive threads/inks.
- ✓ SAR reduction via different techniques.
- ✓ Simulations limited to  **$S_{11}$ , Gain, SAR** in is **local only** (no cloud integration).
- ✓ CST/HFSS.
- ✓ Data collected manually.

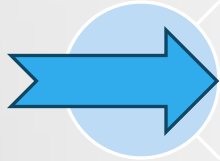
## Wearable Antenna Design Process



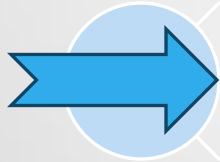
# *Proposed Methodology*



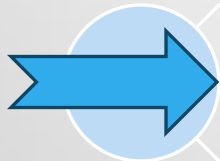
Optimal antenna design and simulation using CST software



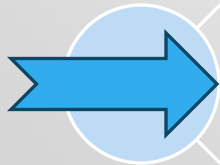
Geometry optimization: patch antenna structure



SAR reduction and gain improvement



Materials: conductive textiles + flexible polymer substrates



Data Flow: Sensors → Antenna → WBAN Hub → Cloud Server → Remote Monitoring

# *How Proposed Project Is Different From Existing Works*

## Existing Projects

High SAR (near unsafe limits)

Narrow bandwidth (single ISM band)

Low efficiency (60–70%)

Only antenna hardware focus

Local measurement only

No real-time access

Mainly academic prototypes

## Proposed Project Idea

$SAR < 1.6 \text{ W/kg}$  (safe for human use)

Wideband ( $\geq 500 \text{ MHz}$ , multiband)

High efficiency ( $> 80\%$ )

Full system (antenna + data collection + cloud)

Remote monitoring via cloud & unique ID

Live monitoring & alerts

Scalable to healthcare, fitness, military

# *PRE-PROCESSING FLOW OF ANTENNA DESIGN*

- ❖ Requirement Analysis
- ❖ Material & Substrate Selection
- ❖ Antenna Geometry Design
- ❖ Pre-Simulation Setup (Boundary, Excitation)
- ❖ Body Phantom Modelling
- ❖ Optimization & Validation

# *Simulation Using CST Studio Suite*

- Antenna model creation in CST
- Define Material Properties
- Setup of boundary conditions and excitation ports
- Assign Frequency Range
- Run Simulation (S11, VSWR, Radiation, SAR)
- SAR & S-parameter analysis (S11, VSWR, Gain)
- Parametric sweeps for optimization
- Export & validation of results

# *Antenna Simulating Parameters*

- **Operating and Electrical Parameters:**
  - ❖ Operating frequency range: **4.17 – 4.25 GHz**
  - ❖ Center frequency: **4.25 GHz**
  - ❖ Application band: **WBAN / Telemedicine**
  - ❖ Reflection coefficient ( $S_{11}$ ):  **$\leq -10$  dB**
  - ❖ Desired impedance matching: **50  $\Omega$**
  - ❖ Voltage Standing Wave Ratio (VSWR):  **$\leq 2$**
  - ❖ Bandwidth defined at  **$S_{11} < -10$  dB**
- **Feed Type**
  - ❖ Feeding technique: **Microstrip line feed**
  - ❖ Characteristic impedance: **50  $\Omega$**
  - ❖ Feed line optimized for impedance matching
  - Feed position adjusted for minimum return loss

- **Radiation and Performance Parameters**

- ❖ Radiation pattern: **2D and 3D**
- ❖ Radiation type: **Semi-directional**
- ❖ Gain: **> 5 dBi**
- ❖ Directivity: **> 6 dBi**
- ❖ Front-to-back ratio: **> 10 dB**
- ❖ Radiation efficiency: **> 70%**
- ❖ Total antenna efficiency: **> 65%**

- **Specific Absorption Rate (SAR)**

- ❖ SAR analysis performed for wearable safety
- ❖ Input power: **1 W**
- ❖ Evaluation frequency: **4.25 GHz**
- ❖ SAR (1 g tissue):  **$\leq 1.6$  W/kg (FCC standard)**
- ❖ SAR (10 g tissue):  **$\leq 2$  W/kg (EU standard)**
- ❖ Human body phantom used for SAR evaluation

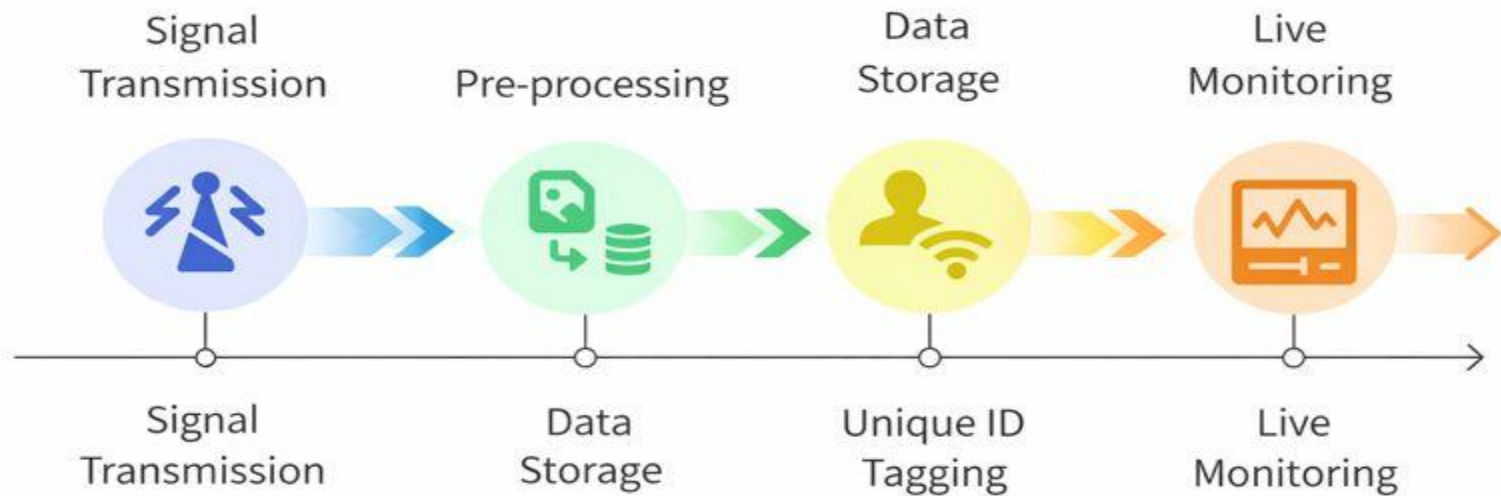
# Cloud Integration (Firebase)

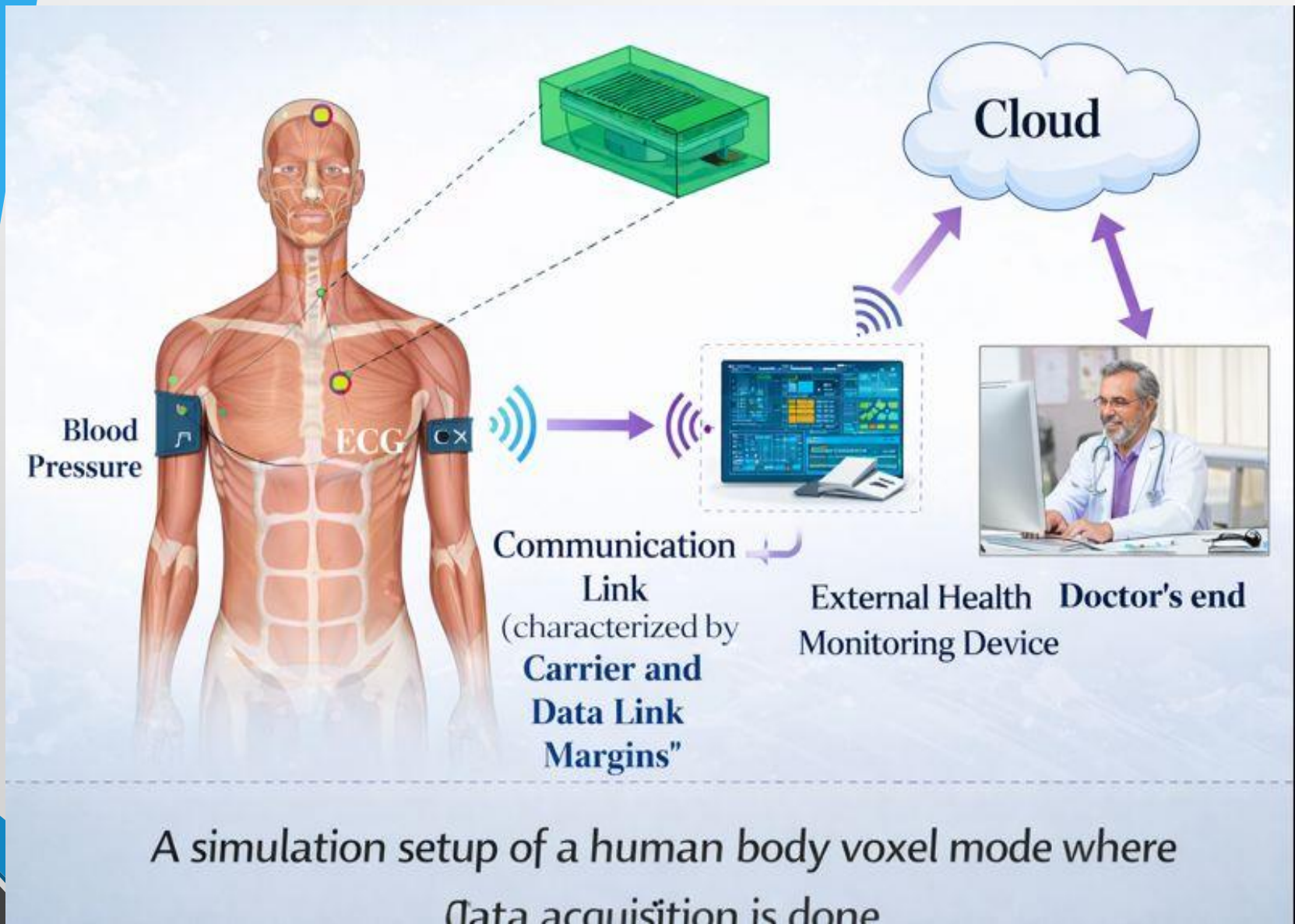


Firebase

- ❖ **Firebase Realtime Database** is used for storing physiological parameters such as heart rate, blood pressure, and temperature in real time.
- ❖ Enables **remote monitoring** of patient data from any location through web or mobile applications.
- ❖ Provides **secure data storage** with authentication and access control.
- ❖ Supports **real-time data updates**, ensuring low latency and reliable cloud communication for telemedicine applications.

**Proposed wearable body area antenna operation flow after acquiring the data**





## *Applications*

- Remote health monitoring & telemedicine.
- Continuous vital tracking (HR, BP, respiration, clots).
- Fitness and sports performance monitoring.
- Military – soldier health status. monitoring and tracking

## *Advantages*

- Lightweight & flexible – comfortable wear.
- Safe: Low SAR, minimal radiation.
- Portable cloud storage with unique ID access.
- Real-time monitoring with IoT/cloud integration.
- Recorded data can be accessed from any location

## **REFERENCES:**

- High-Gain Low-SAR Wearable Antenna Based on AMC in WBAN [!\[\]\(df97338b4ed9c84de0aa8b40b09055fa\_img.jpg\) Link](#)
- Flexible and Wearable Antenna for Biomedical Applications [!\[\]\(b22c7e8560907ffb4e97a8ca6af278e9\_img.jpg\) Link](#)
- Miniature Implantable and On-Body Antennas for WBAN [!\[\]\(0a0ac6cc6d55a8d77e6e348c940ad042\_img.jpg\) Link](#)
- Design, Analysis and Applications of Wearable Antennas – Review [!\[\]\(a194910639352cc7e6ba1807c01ec960\_img.jpg\) Link](#)
- Impact of Hub Location on WBAN Performance for Fitness Applications [!\[\]\(39e21277690d79d1c52b3a86fa83368d\_img.jpg\) Link](#)
- Biosensors in Wearable Healthcare Devices [!\[\]\(2b9dc98ffaf75730ad126d509d400942\_img.jpg\) Link](#)



THANK YOU